
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their impressive performance, generative image models trained on large-scale datasets frequently fail to produce images with seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that are reasonably expected to appear in the training data. To date, these failure modes have been documented only anecdotally, leaving unresolved a central quantitative question: are they idiosyncratic anomalies, or measurable, systematic limitations of these models? To answer this quantitatively, we introduce a systematic approach for identifying and measuring “conceptual blindspots”—concepts present in the training data but absent or misrepresented in a model’s generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings and, crucially, to assign each concept a single scalar energy-difference score that directly quantifies its over- or under-representation in real versus generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with a dictionary of 32,000 concepts—to our knowledge the largest such SAE to date—which yields the concept-level granularity needed to measure conceptual disparities precisely. Applied head-to-head to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our measure reveals a consistent and quantifiable pattern: all four models are more likely to suppress concepts than to exaggerate them, and the same score pinpoints specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the same score further isolates memorization artifacts—instances where models reproduce highly specific visual templates seen during training. Overall, we contribute a theoretically grounded and quantitatively operationalized framework that measures a generative model’s conceptual fidelity with respect to the underlying data-generating process, turning previously anecdotal failure reports into reproducible, comparable numbers.

1 Introduction

Generative image models trained on large scale datasets have achieved unprecedented capabilities, allowing their use in applications both within the vision domain [1–9] and well beyond that [9–15]. Despite this success, a growing body of qualitative [16–18] and quantitative studies [19, 20] consistently reports that, at times, models struggle to generate images with relatively simple concepts, e.g., human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. In fact, when prompted to generate images containing such concepts, models tend to produce outputs with related structures, but not precisely the ground truth concept—e.g., producing hands with six fingers. These failure modes, which we call “conceptual



Figure 1: Our single scalar score $\delta(k)$ quantifies, on a $[0, 1]$ scale, a model’s tendency to over- or under-generate a concept c_k relative to its natural-data frequency: values below the natural-frequency midpoint indicate suppression and values above it indicate exaggeration. Using fixed thresholds, we label concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The depicted images, generated by four popular generative image models, contrast concepts flagged at these extremes against concepts the models render faithfully.

blindsports¹ can be unintuitive, since one may reasonably expect models have had enough exposure to demonstrations accurately detailing such concepts. The recurring pattern across these reports raises a question that has so far resisted measurement: do such failures reflect intriguing quirks of a few specific concepts, or are they the visible surface of a broader, quantifiable phenomenon under which, across a wide spectrum of concepts, models are systematically too unlikely—or too likely—to produce images containing them?

Answering this question quantitatively remains infeasible with existing approaches for evaluating generative image models [28, 29], each of which measures something other than concept-level fidelity. Specifically, the most widely used measures are coarse-grained scalars meant to assess image realism, e.g., FID [30]; because they aggregate over whole images, they report a single realism number and are blind to which concepts drive a distributional failure. Methods like CLIPScore instead quantify generation diversity or distribution-coverage statistics [31–33], which speak to our question only in aggregate and never at the granularity of individual features or concepts [34, 35], leaving specific conceptual blindspots unmeasured. Finally, qualitative analyses for evaluating generative models, such as participant surveys [5, 17, 36–38] or open-ended exploration [39], can surface individual failures but yield no reproducible, per-concept quantity that can be recomputed across model families and compared on a common scale.

This work. Motivated by the above, we argue that identifying and analyzing conceptual blindspots in a generative image model requires a methodology that, in an automated and unsupervised manner, assigns each concept in the data distribution a measurable score capturing the mismatch between its odds of generation under the true data-generating process versus under the trained model. Our contributions in this work are three-fold:

- **Formalizing and Quantifying Conceptual Blindspots in Generative Image Models.** We introduce a systematic framework that reduces the presence of a conceptual blindspot to a single, bounded, per-concept score comparing a model’s generations to natural images (Section 3). This formalization replaces anecdotal or human-defined concept evaluations with a principled, reproducible number that ranks a model’s limitations.
- **A Scalable, Unsupervised Approach for Measuring Conceptual Blindspots using Sparse Autoencoders.** We develop a structured methodology using sparse autoencoders (SAEs) to extract and directly compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with a dictionary of 32,000 concepts—the largest such RA-SAE to date—which sets the resolution at which we can measure conceptual disparities.
- **Exploratory Tool and Empirical Analysis.** Our exploratory web tool enables both distribution- and datapoint-level measurement of blindspots across models (Sections 5.1, 5.4). Applying our method head-to-head to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we find a consistent, model-wide bias toward suppression over exaggeration, and we pinpoint specific suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), shown in Fig. 1.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.

77 The model, code, and web tool are available at a software repository.

78 2 Related Work

79 2.1 Explainability in Vision

80 Early work in explainable AI, including computer vision, focused on methods for attribution of
81 influential input regions [40, 42]. However, these methods offered limited semantic information
82 about learned representations and, under controlled tests, often produced explanations no better than
83 random [43, 45]. To address these issues, concept-based interpretability [46] emerged to identify
84 semantically meaningful directions in neural networks, revealing not just where they look but what
85 concepts and structures they employ [47, 49].

86 Recent work demonstrates that popular concept-based interpretability methods—ACE [44],
87 CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—essentially address the same dictionary
88 learning task under different constraints [52]. Among these approaches, sparse autoencoders (SAEs)
89 stand out as the most scalable for concept-based interpretability, making them the natural substrate
90 for a large-dictionary, concept-level measure. While recent studies quantify limitations of the orig-
91 inal SAEs—including overly specific features [53], compositionality challenges [54], and limited
92 intervention effects [55]—improved SAE variants have emerged, including the archetypal SAE
93 (RA-SAE) [56] that we build on.

94 Beyond SAEs, other interpretability methods include prompt-based probing [57], attention map or
95 activation visualizations [58, 59], and dataset-level statistics [32, 33] (e.g., diversity or distribution
96 coverage metrics); each yields only a partial, aggregate signal for the questions we pose. Crucially,
97 these measures report numbers at the level of full images or prompts rather than at the granularity of
98 fine-grained features and concepts [34, 35], so they cannot localize a discrepancy to a specific concept.
99 Furthermore, they depend on subjective interpretation and do not distinguish between distinct failure
100 modes [60]. As a result, existing methods and metrics fall short of a reproducible, concept-level
101 quantification of where generative image models are weak [28].

102 2.2 Generative Image Models

103 Diffusion-based methods have become dominant across various modalities in generative vision
104 modeling, including image [4, 61–64], video [65–72], and 3D [7, 73–75]. In the domain of image
105 generation, this can be traced back to denoising diffusion probabilistic models (DDPMs) [76], which
106 were later extended to non-Markov diffusion processes with denoising diffusion implicit models
107 (DDIMs) [77].

108 The Stable Diffusion (SD) [9] model family made DDPMs highly accessible both in the research and
109 open-source communities. The original SD was followed by several subsequent versions, including
110 SD 2 [78], SD 3 [79], SD XL [80]. Many modifications and extensions of the SD architecture have
111 emerged, enabling additional constraints for the diffusion process (e.g., style [81, 82], pose [83],
112 and identity [84, 85]) as well as different input modalities, such as image-to-image generation.
113 Different Latent Diffusion Models built on top of SD—including Kandinsky [86], PixArt [87],
114 DeepFloyd [4, 88], and FLUX [89]—have also emerged.

115 2.3 Datasets for Generative Image Models

116 The recent success of generative vision models is largely attributed to the abundance of computational
117 resources and large-scale internet datasets [90–92]. Specifically, LAION-5B [93] has played a
118 key role in the training of open-source text-to-image models, including SD and its derivatives.
119 Scraped from Common Crawl [94], this dataset spans over 5B image-caption pairs—2.3B of them in
120 English—dwarfing other prominent corpora such as COYO-700M [95] (700M pairs) and Conceptual
121 Captions [96] (3M pairs) by roughly one to three orders of magnitude. This scale makes the recurrence
122 of simple-concept failures all the more striking, since the relevant concepts are almost certainly
123 present in the training data.

124 As LAION-5B gained popularity, concerns grew over its biases [97–101]. Despite filtering attempts,
125 harmful content persisted [97–99], including NSFW material [100] and hundreds of CSAM in-
126 stances [101], prompting its temporary removal from official channels. The dataset also contains

many low-quality images [102] and internet-style captions (e.g., product descriptions) that misalign with how users prompt trained models [103]—distributional properties that, as we later quantify, correlate with concept-level generation failures.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks to find sparse representations of input data, where each sample can be reconstructed using a linear combination of few dictionary atoms [104–106]. Built upon compressed sensing theory [107–108], the field evolved from early vector quantization methods [109] to sophisticated approaches including Non-negative Matrix Factorization [110–111], Sparse PCA [112], and K-SVD [113]. Recent advances include online methods [114], structured sparsity [115], and theoretical guarantees [116–117], alongside growing connections to deep learning [118–119].

Advances in sparse coding have also been leveraged to study the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was deployed to earlier generative architectures through concept-bottleneck models, which require human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both approaches, however, are bounded by a fixed set of human-defined concepts and therefore cannot quantify the many trends a user does not explicitly register—the very trends an unsupervised, 32,000-concept measure is designed to surface.

Aggregate metrics (e.g., precision, recall, density, and coverage [35, 123]) and latent density scores, which predict sample quality based on the model’s latent space [124], have emerged to evaluate generative image model capabilities. While these effectively quantify the overall size of a distributional gap, they report a single aggregate number and cannot attribute that gap to specific concepts that are under- or over-represented.

Together, these limitations motivate a scalable, unsupervised framework that assigns each concept its own measurable failure score, enabling systematic and comparable quantification of concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing the notion of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. The purpose of this formulation is quantitative: it lets us collapse each such discrepancy into a single, principled, bounded score that measures how far a concept is under- or over-represented by a generative model relative to its data distribution. The process is illustrated in Fig. 2. While we rely on standard assumptions in this pursuit [126–130], we find empirically that meaningful, measurable phenomenology is elicited even when these assumptions are violated.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula.

In essence, the individual dimensions of the latent space reflect the **Concepts** underlying the data-distribution $P_{\mathcal{X}}$, defined over some observation space of images $\mathcal{X}$. For example, different latents may correspond to concepts like color, shape, size, location, and so on [131–132]. We also let the data-generating process associate a text-description $t \in \mathcal{T}$ with any image sampled from the data distribution, but do not explicitly model it. These text descriptions can then be used to train a text-conditioned **Generative image model** g_θ , with parameters

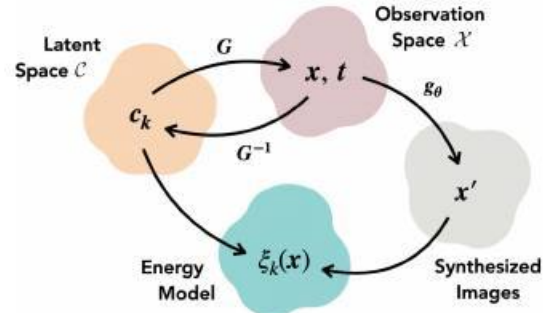


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP. S_θ generates images x' based on t . The energy model $\xi_k(x)$ extracts a per-concept energy from both x and x' ; contrasting these two energies yields the scalar

180 θ , on a set of image-text pairs to map a noise
 181 signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a text-description of
 182 the scene t to produce images x illustrating the
 183 latter.

184 To define conceptual blindspots in the model g_θ ,
 185 we must assess the probability mass assigned to
 186 a concept by the data-generating process, com-
 187 paring it to the mass assigned by the model. To
 188 this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and define a set of syn-
 189 thetically generated images $\mathcal{D}'_\mathcal{X}$ using the text descriptions. To estimate the probability of occurrence
 190 of a concept, we consider an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional
 191 vector, where each dimension associates a scalar representing the energy in the k^{th} concept, i.e.,
 192 $\xi_k(x) = E(c_k)$. These estimates are feasible because we assume the data-generating process is
 193 invertible. Correspondingly, the probability the data-generating process associates with the image
 194 x can then be defined as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (where Z_ξ is the partition function), hence
 195 yielding a population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{x \in \mathcal{D}_\mathcal{X}} p_k(x)$. Using this and the sigmoid map $\sigma(\cdot)$,
 196 we define below the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$.

197 **Definition 2 (Energy Difference).** *Let $x \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-*
 198 *generating process G , and let $x' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let*
 199 *$\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the*
 200 *energy difference for concept k as:*

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{x'}[\xi_k(x')] - \mathbb{E}_x[\xi_k(x)]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

201 *where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{x \in \mathcal{D}} \xi_k(x))$*
 202 *denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .*

203 Concretely, the energy difference in the k^{th} concept is the fraction of a concept’s total probability mass
 204 that falls on the generated set $\mathcal{D}'_\mathcal{X}$ rather than the natural baseline $\mathcal{D}_\mathcal{X}$; it is bounded in $[0, 1]$, equals
 205 0.5 when the two match exactly, and moves toward 0 or 1 as the model increasingly suppresses or
 206 exaggerates the concept. This single bounded number is the measure on which our entire quantitative
 207 analysis rests. Based on it, we now define conceptual blindspots as follows.

208 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** *Given a generative image model*
 209 *g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in*
 210 *the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.*

211 Overall, we define a conceptual blindspot as a concept whose score deviates markedly from the
 212 balanced value of 0.5, either through suppression or exaggeration, relative to its prevalence under the
 213 data-generating process. Suppressed concepts fall into the extreme low tail ($\delta(k) < \lambda_{\min}$), whereas
 214 exaggerated concepts fall into the extreme high tail ($\delta(k) > \lambda_{\max}$). Throughout our analysis, we fix
 215 the thresholds at $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$, so that a flagged concept has at least a 9:1 imbalance
 216 in probability mass between the two distributions—a deliberately conservative bar that isolates only
 217 the most pronounced discrepancies.

218 We also note this definition is related to the idea of “mode collapse” studied in past work (e.g., see
 219 Bau et al. [39]): the key difference is the granularity at which the odds are quantified. Mode collapse
 220 measures exaggerated or suppressed odds of generating *entire images*, whereas our score measures
 221 the shifted odds of each specific concept, one dimension at a time. For example, if a model fails
 222 to produce images of an object with a white background, we say this concept is a suppressed
 223 conceptual blindspot.

224 4 Method: Operationalizing the Definition of Conceptual Blindspots

225 We next describe the pipeline that turns the definition above into a computable, per-concept number
 226 for a generative model g_θ . As per Sec. 3, computing the score requires two ingredients: (i) a set
 227 of images sampled from g_θ that can be compared, prompt-for-prompt, against the ground-truth

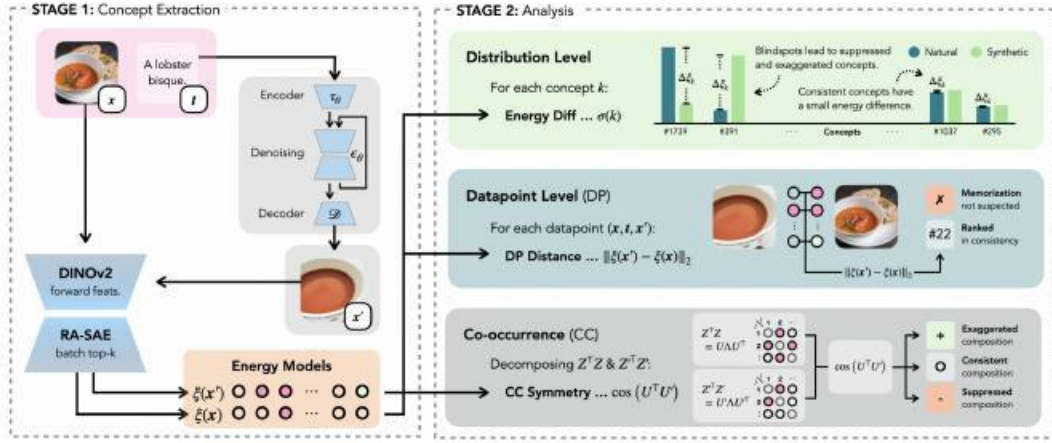


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , the concepts in x and t are extracted by obtaining each image’s DINOv2 features, which a RA-SAE then maps into sparse concept embeddings, yielding the energy vectors $\xi(x)$ and $\xi(x')$. Each coordinate is directly comparable: $\xi_k(x) = E(c_k)$ is the energy in the k^{th} concept, and the paired real-vs-generated energies feed straight into the scalar score $\delta(k)$.

generative process, and (ii) an energy model that supplies the per-concept energies entering that comparison. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$ to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in a model g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs (x, t) sampled from the data-generating process $\mathcal{G}$ and their synthetic counterparts sampled from the generative model g_θ using the text descriptions.

Specifically, given t , we synthesize a counterpart image x' using a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling occurs in latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \varepsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded via a pretrained VAE to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper, we treat g_θ as a black box that maps prompts to images: $t \mapsto x'$.

Defining the Energy Model. Building on prior work that shows the ability of self-supervised learning methods to invert the data-generating process and identify the energy function underlying it up to linear transformations [126, 128, 133, 134], we use DINOv2 [135]—a state-of-the-art self-supervised model—for our analysis. Under the expectation that the number of concepts underlying the DGP is larger than the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) on its features to identify subspaces corresponding to these concepts [50, 56, 137–139]. The intuition here is that if the concepts underlying the generative process are modeled via approximately orthogonal directions by DINOv2 (as assumed in our independence constraint in Def. 1), then an SAE should be able to isolate these concepts along individual dimensions in its latent space [136]. The activation associated by the SAE to a dimension will serve as our approximation of the ground-truth energy function assigned to the concept modeled by that dimension.

Formally, using $f : \mathcal{X} \rightarrow \mathbb{R}^d$ to denote our feature extraction module (i.e., the DINOv2 model), we extract features $\mathbf{a} = f(x) \in \mathbb{R}^d$ from both natural and synthetic images from datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$. Assuming the count of image-text pairs is n , we stack the real and generated features into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. We then decompose each feature vector into a sparse combination of learned concept atoms using an SAE. Specifically, let $\mathbf{D} \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the matrix of activations $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, where each row $\mathbf{z}_i = \Psi(\mathbf{a}_i)$ represents the concept decomposition of an image. The decoder reconstructs features via $\mathbf{Z}\mathbf{D}^\top$, and the SAE is

trained to minimize the reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs often drift toward arbitrarily oriented dictionaries, making downstream numbers highly sensitive to the random seed—an unacceptable property for a measure meant to be compared across models. To remove this seed-dependence and make our scores reproducible, we employ the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to be a convex combination of training data. Specifically, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Thus every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and any reconstruction $ZD^\top$ resides inside the conic hull of the data $\text{cone}(A)$. This ensures learned concepts remain faithful to the support of the data distribution [56]. Once trained, the SAE provides a consistent set of sparse codes: Z for real images and Z' for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of the concept serving as energy values for our analysis of conceptual blindspots.* In summary, our method defines a structured pipeline that, given a prompt and its associated real image (t, x) , produces two sparse concept vectors (z, z') in a common basis, so that every one of the 32,000 concept coordinates can be compared directly and reduced to its scalar score.

To restate the full pipeline: starting from a captioned image (t, x) , we synthesize a counterpart x' and map both images into a shared, sparse concept space via a vision encoder and a trained SAE, yielding interpretable representations (z, z') that furnish the per-concept energies from which every quantitative result in Section 5 is computed.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—on a common benchmark of $|\mathcal{X}| = 10,000$ image-text pairs and their prompt-matched generations (Fig. 4), so that every model is scored against the identical natural baseline. We report results at three complementary levels of granularity (Fig. 3): a ● **distribution-level** evaluation quantifies the balance of suppressed versus exaggerated concepts; a ● **datapoint-level** analysis ranks individual pairs to isolate failures tied to ambiguity, omission, and memorization; and a ● **compositional** analysis measures distortions in concept co-occurrence geometry.

A central contribution is an interactive exploratory tool, shown in Fig. 5. The tool is a web-based interface built around a UMAP projection of concept representations, enabling side-by-side visualization and comparison of concept-level energy differences. It is publicly available at a software repository, together with the pre-computed energy-difference scores for all 32,000 concepts across the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). Every quantitative finding below is derived from these scores. Its primary functionalities, which support these analyses, include:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization spanning all 32,000 concepts from the RA-SAE. Each scatter point is one concept, color-coded by its energy-difference score, so cross-model differences in the score are immediately legible.
- **Inspect concepts.** Each concept has a card with its quantitative statistics, representative real and generated images (x, x') , and visualized co-occurrence patterns.



Figure 4: **Representative image-caption pairs** from LAION-5B, each shown alongside prompt-matched generations from all four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). This one-to-one pairing is what makes the per-concept comparison strictly controlled: every model receives the same caption and is scored against the same natural image.

309 • **Explore blindspots.** Beyond the UMAP and per-concept views, the tool exposes global
 310 rankings of concepts by energy difference, surfacing the most strongly suppressed and
 311 exaggerated blindspots at the extremes of the score.

312 Rather than exhaustively studying one phenomenon, we report a set of high-level quantitative findings
 313 that demonstrate the measure’s versatility and enable broader, customizable exploration.

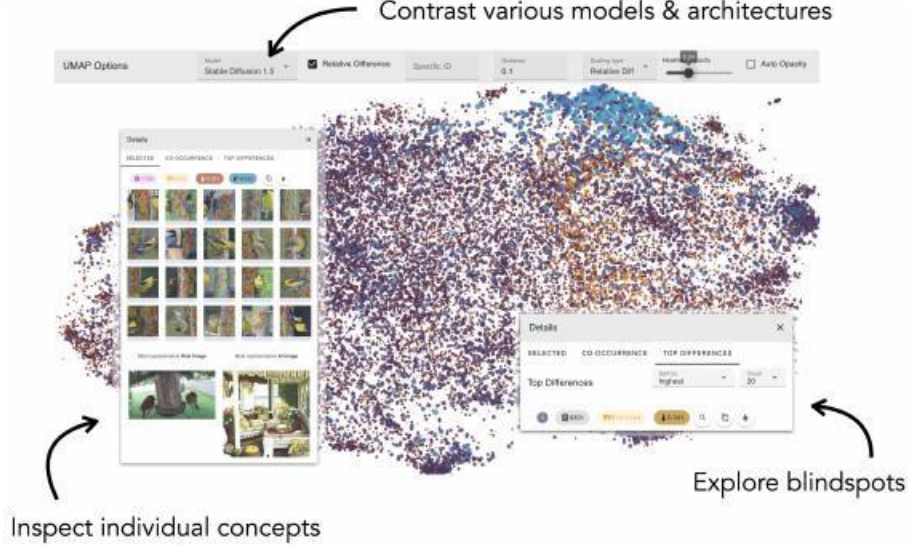


Figure 5: **Overview of the Exploratory Tool.** The web interface displays a UMAP projection for each evaluated model, where each dot is one concept, color-coded by its energy-difference score. Selecting a concept opens a detail panel with its quantitative statistics and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences is shown alongside global rankings of concepts by energy difference, placing the strongest blindspots at the top.

314 5.1 Generative Image Models Cannot Produce Unseen Concepts

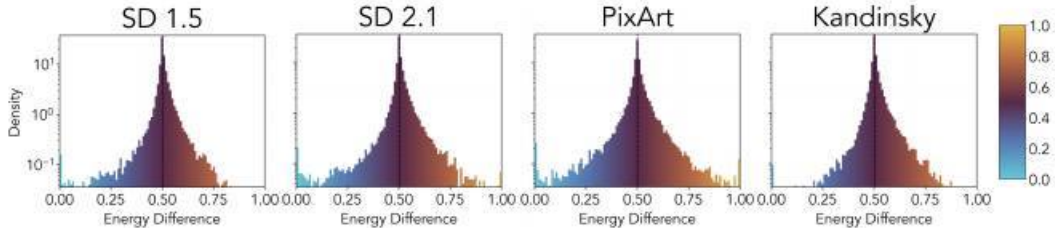


Figure 6: **Concept Energy Distribution.** Log-scale histograms of the energy difference $\delta(k)$ over all 32,000 concepts, comparing the natural and synthesized distributions for each evaluated model. Mass left of center marks suppressed (under-represented) concepts; mass right of center marks exaggerated (over-represented) concepts. In every one of the four models the left tail is visibly heavier and longer than the right, so the dominant, consistent pattern is suppression rather than exaggeration.

315 To quantify the disparity between each generative model and the natural image distribution, we first
 316 examine the marginal energy difference $\delta(k)$ over all 32,000 concepts learned by the RA-SAE. As
 317 defined in Sec. 4, this quantity measures the relative prevalence of each concept in the synthesized
 318 versus natural image sets: $\delta(k) < 0.1$ flags concept k as under-represented (suppressed), while
 319 $\delta(k) > 0.9$ flags it as over-represented (exaggerated). Fig. 6 plots the full distribution of $\delta(k)$ for
 320 each of the four evaluated models. Two patterns hold across every model. First, all four histograms
 321 are heavy-tailed with substantial mass at both extremes, evidence of systematic—not incidental—
 322 discrepancies in concept coverage. Second, and more tellingly, the left (suppressed) tail is uniformly
 323 denser and longer than the right (exaggerated) tail, establishing suppression as the dominant failure

direction. This asymmetry is captured quantitatively by the negative skewness of every distribution, ranked here from most to least skewed: SD 2.1 is the most left-skewed at -0.54 , followed by SD 1.5 and PixArt (both -0.40), with Kandinsky the least skewed at -0.23 . The sign is negative in all four cases, so the suppression bias is universal; its magnitude varies by up to $2.3\times$ across models.

All four models exhibit both suppressed and exaggerated concepts, yet their profiles differ in measurable ways. PixArt, for example, has the widest spread, consistent with a more strongly suppressed concept distribution. Despite these per-model differences, the consistently negative skew across all four distributions makes concept omission the common, quantifiable tendency; we dissect its structure and drivers in the sections that follow.

5.2 Structure and Specificity of Conceptual Blindspots

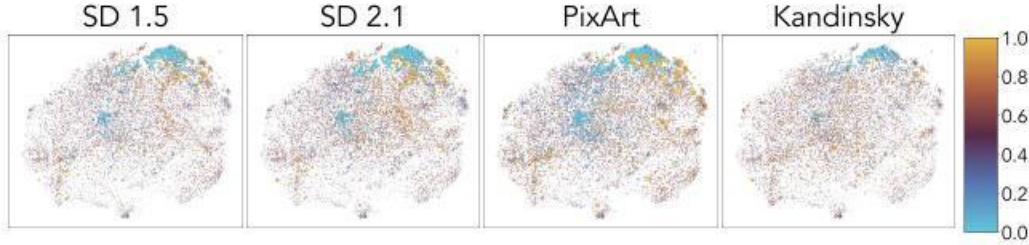


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of all 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions (suppressed concepts at the blue end, exaggerated at the yellow end). Blindspots are not scattered at random: they form contiguous, same-colored clusters in every model, showing that the score varies systematically over concept space rather than as noise.

Having quantified the marginal discrepancies in concept frequency, we now ask whether they are structured. We embed the full set of 32,000 concepts into two dimensions with UMAP on the sparse codes and color each concept by its $\delta(\cdot)$ score. As Fig. 7 shows, distinct clusters emerge in every model. These clusters coincide with contiguous regions of conceptual blindspots—most sharply for suppressed (blue) concepts—demonstrating that blindspots are far from random: they are spatially organized, pointing to shared biases in either the training distribution or architectural priors. To measure how consistent these blindspot patterns are across models, we correlate the concept-level $\delta(\cdot)$ scores between model pairs. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vector of SD 1.5 and those of every other model. The numbers separate cleanly into two regimes. SD 1.5 and SD 2.1—which share an architecture and training pipeline—are strongly correlated at $r = 0.82$. By contrast, SD 1.5 correlates only weakly with PixArt ($r = 0.41$) and Kandinsky ($r = 0.46$): roughly half the strength of the same-family pair. In other words, changing the architecture nearly halves the agreement in blindspots, quantifying how much of a model’s blindspot profile is architecture-specific rather than dataset-wide.

Taken together, the correlations above quantify a clear split: some blindspots are shared across all models—likely inherited from the dataset—while others are strongly model-specific, as evidenced by

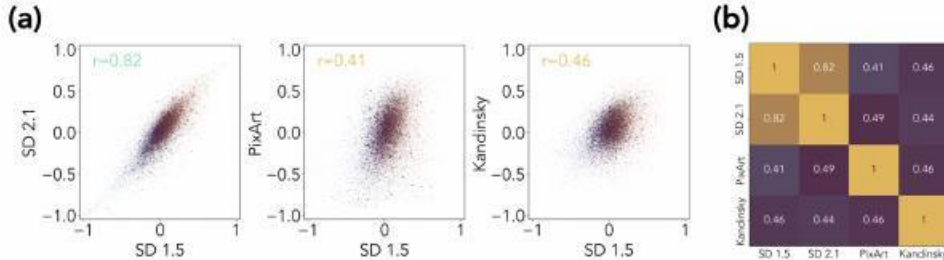


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with the Pearson correlation reported top-left of each panel. The same-family pair SD 1.5–SD 2.1 aligns strongly ($r = 0.82$), whereas cross-architecture pairs correlate roughly half as strongly ($r = 0.41$ – 0.46), quantifying the extent of model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** Heatmap of the pairwise Pearson correlations for δ over all models, summarizing at a glance how much of each blindspot profile is shared versus model-specific.

the drop from $r = 0.82$ within a family to $r \approx 0.4$ across architectures. This measured gap motivates studying both the shared and the model-unique blindspots separately in the sections that follow.

5.3 Qualitative Examples of Blindspots

To ground the score, we next inspect concepts sitting at its extremes, contrasting a blindspot shared by all models against one confined to a single model. In Fig. 9a we show a concept suppressed by all four models: none reproduce solid white on documents. Despite the caption explicitly naming this concept, not one of the generated images realizes it—a unanimous failure that matches the low δ score assigned to this concept in every model and confirms this region of concept space is systematically under-sampled. In direct contrast, Fig. 9b isolates a model-specific blindspot: three of the four models render pan faithfully, yet it is conspicuously absent from Kandinsky’s generations alone. This single-model outlier is exactly the behavior predicted by the low cross-architecture correlations of Sec. 5.2 where agreement was high within a family but only $r \approx 0.4$ across architectures.

Turning to the opposite tail of the score, Fig. 10 presents a clear exaggeration: the concept shadow under animal is over-emphasized in the generated images. While shadows are plausible in principle, their consistently pronounced rendering across models—compared with the more subtle and variable shadows in natural images—is the visual signature of the high δ score this concept receives, indicating an overactive prior. Notably, our search for concepts exaggerated by only a single model turned up no clear examples, mirroring the histogram finding that the exaggeration tail is both thinner and more uniform across models: exaggerations, unlike suppressions, appear to be approximately universal.

Together, these examples show concretely how the two ends of the δ scale translate into visible over- and under-generation, confirming that our energy-based measure surfaces both shared and model-specific failure modes. It even quantifies surprising limitations—such as the unanimous failure to reproduce clear or solid background elements like whitespaces on documents—whose universality points to architectural constraints or training-data biases that transcend individual model idiosyncrasies.

Informative as these aggregate scores are, they raise a sharper question: do blindspots exist only in the population average, or are they measurable pair-by-pair at the



Figure 10: **Example of an Exaggerated Conceptual Blindspot.** Four synthesized images $\mathcal{D}'_{\chi}$, archetypal for the concept shadow under animal, alongside the corresponding natural image $\mathcal{D}_{\chi}$ and caption t used to generate it. The ac-

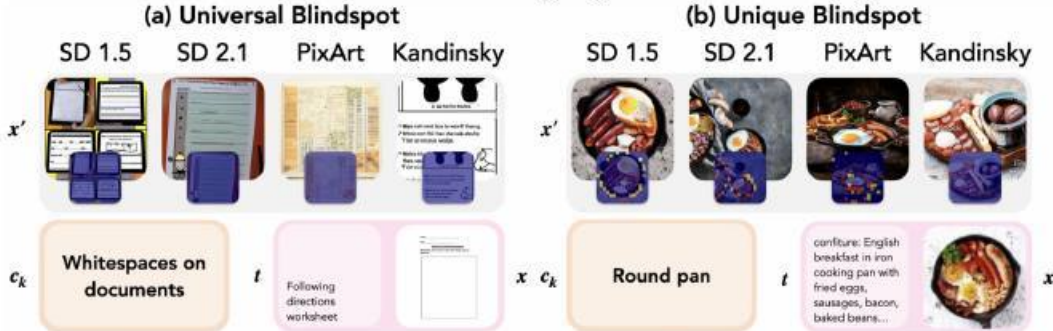


Figure 9: **Examples of Suppressed Conceptual Blindspots.** The natural image x , representative c_k and t , shown alongside four synthesized images x' generated using S_{θ} . Top: a *universal* blindspot, suppressed in all four models. Bottom: a *model-unique* blindspot, suppressed only in Kandinsky while the other three render it faithfully—the two ends of the shared-vs-specific split quantified in Sec. 5.2

level of individual dat-
apoints? Answering it
lets us attribute the score
to concrete mechanisms—
prompt misinterpretation, la-
tent memorization, or
both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we rank individual natural-vs-generated image pairs by their $\delta(\cdot)$ averaged across all concepts and inspect the two extremes of that ranking—the pairs with the smallest and the largest divergence. Sorting by the score cleanly separates two opposite regimes of model behavior, both of which we find informative. At the low-score end, Fig. 11a shows pairs with near-zero average $\delta(\cdot)$. Here the generated images are conceptually near-indistinguishable from the originals. Yet qualitative inspection reveals this near-perfect score is achieved not through faithful prompt-following but through near-verbatim replication of memorized templates: the repetitive visual structures (e.g., identical clothing outlines or object arrangements) indicate copying of over-frequent training patterns. A near-zero score, in other words, can be a symptom of memorization rather than of genuine fidelity. At the high-score end, Fig. 11b shows the pairs with the largest $\delta(\cdot)$. These mark severe conceptual divergence between the synthesized and natural image. Some of this divergence traces to underspecified or noisy captions, but the remainder exposes genuine blindspots: the prompt names a clear concept that is plainly present in $\mathcal{D}_X$, yet the model fails to realize it in $\mathcal{D}'_X$. Such cases show that even with adequate language grounding, certain concepts lie beyond the model’s generative reach—and the score localizes exactly which pairs those are.



Figure 11: • **Datapoint-level Conceptual Alignment.** (a) The lowest-scoring pairs (near-zero energy difference), where near-perfect alignment coincides with apparent memorization of training patterns. (b) The highest-scoring pairs (large energy difference), where the score flags substantial concept divergence traceable to prompt ambiguity or genuine model limitations. Reading the ranking from either end thus isolates a distinct failure mode.

5.5 Analyzing Post-Training Effects

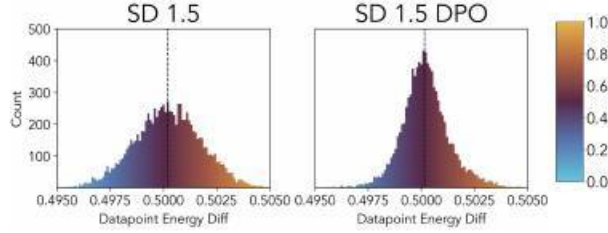


Figure 12: **Effect of DPO on Concept Fidelity.** Overlaid histograms of datapoint-wise energy differences between the synthesized and natural distributions for SD 1.5 with and without DPO. The DPO variant’s distribution is shifted toward lower values and is visibly narrower, i.e., both a smaller median divergence and a tighter spread—a direct, quantitative improvement in conceptual alignment.

Post-training protocols, e.g., safety fine-tuning, have been argued to reduce the diversity of model generations [140]. Because our pipeline yields a single scalar per image pair measuring its divergence from the ground-truth DGP, it is well suited to quantify the effect of DPO—a popular safety fine-tuning protocol [141]—in a controlled A/B comparison. We compare two variants of SD 1.5 that differ only in DPO training: one with, one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$, giving a per-pair divergence score. Fig. 12 overlays the histograms of these datapoint-wise scores for the two variants. The result is unambiguous and points in a single direction: the DPO variant shifts the entire distribution toward lower divergence, lowering both its median and its spread, so its generations are on average *and* more consistently closer to the ground-truth DGP. This is quantitative evidence that DPO regularizes the model’s concept distribution, pulling outputs toward the semantic content of the seen inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, the measured reduction in divergence shows that its human-preference objective indirectly improves the match to the training distribution. Critically, the improvement is two-sided: DPO reduces *both* over- and under-activation of individual concepts relative to the natural baseline, rather than trading one off against the other. These findings demonstrate that our pipeline can quantify the downstream effect of a post-training intervention not just as a change in output quality, but as a measurable reshaping of the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

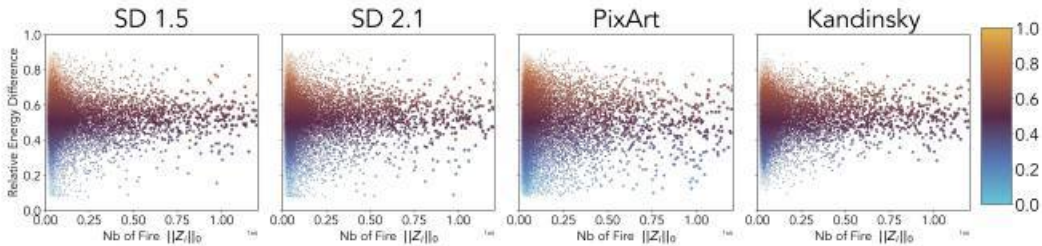


Figure 13: **Concept Fidelity Across the Frequency Spectrum.** Per-model scatter plots relating a concept’s empirical frequency in natural data (x-axis) to its energy difference (y-axis); each point is a concept, sized by its activation frequency. The same trend recurs in all four models: high-frequency concepts cluster near balanced fidelity, while rare concepts fan out toward large discrepancies—so frequency is a consistent predictor of blindspot severity.

We previously hypothesized that conceptual blindspots are not merely architectural artifacts but may follow directly from distributional peculiarities of certain features. Here we test this hypothesis quantitatively by asking whether concepts rarely activated in natural images—those in the long tail of the data distribution—are precisely the ones that become blindspots. We process the natural dataset $\mathcal{D}_X$ through the trained SAE and, for each concept k , compute its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, the count of non-zero activations of concept i across all images. We then relate this frequency to the absolute energy difference observed in the generated outputs. Fig. 13 plots this relationship for all

four models, and the pattern is consistent across every one of them: frequent concepts concentrate at low energy discrepancy, whereas rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit the largest alignment errors. The score therefore covaries systematically with frequency, indicating that many blindspots are not random model quirks but a measurable consequence of long-tail concept distributions. This points to remedies beyond architecture—e.g., data reweighting or augmentation—targeted at the low-frequency regime.

6 Discussion

Across four popular generative image models, a single per-concept score surfaced a consistent set of quantitative findings: a universal suppression bias (negative skew in all four models, from -0.23 to -0.54), a sharp same-family versus cross-architecture split in blindspot agreement ($r = 0.82$ vs. $r \approx 0.4$), a two-sided reduction in divergence under DPO, and a systematic link between low natural-data frequency and blindspot severity. Each of these could warrant its own dedicated study; rather than exhausting any one, we use them to demonstrate the versatility of the underlying measure and tool. Out of the box, they quantify which concepts a model struggles to generate, detect memorization artifacts, flag datapoints with insufficient captions, measure post-training effects, and compare conceptual shifts across architectures on a common scale. This opens space for follow-up work to deepen each analysis, widen the set of evaluated architectures, and trace the root causes of blindspots. Future work could also explore hierarchical representations of concepts in RA-SAE for a more fine-grained measurement.

Limitations. We highlight several limitations. Because concept extraction relies on DINOv2 and RA-SAE, our measure can only score concepts these models represent; concepts they capture poorly fall outside its reach. And while our benchmark of 10,000 image-text pairs is substantial—large enough to estimate each $\delta(k)$ tightly—it may still under-sample the long tail of rare concepts, concept co-occurrences, and other compositional statistics.

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Appendix

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A Experimental Setup

This section details the experimental setup behind our head-to-head analysis of four popular generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or its subsets/derivatives and, crucially, all evaluated on the identical set of prompts and natural images so that their scores are directly comparable. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face datasets library, (2) validating proper URL structure and resource availability via HTTP HEAD requests, and (3) random sampling with replacement until reaching the target count of 10,000 valid samples. This yields the observation set D_G of (x, t) tuples—the common natural baseline against which all four models are scored—while ensuring every pair remains accessible despite a large fraction of the original image URLs having gone offline since release. Additional examples of synthesized images are shown in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in strict one-to-one correspondence with D_G , yielding triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G we reuse the same caption t to synthesize a counterpart image x' from each model g_θ . This prompt-matched design is what makes the per-concept comparison a controlled one: any difference in $\delta(k)$ is attributable to the model, not to differing prompts or reference images.

The synthesis follows each architecture’s standard text-to-image pipeline, implemented with the Hugging Face `diffusers` library and models loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters, holding output resolution fixed across models.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/ruwnayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. For each evaluated model, we compute the energy difference $\delta(\cdot)$ for all 32,000 concepts, applying a sigmoid transformation with temperature $T = 0.8$ during normalization. The

resulting scores are summarized as log-scale density histograms with 100 bins spanning the full $[0, 1]$ range of the score, from which the per-model skewness values reported in the main text are computed.

Section 5.2 We embed the complete set of 32,000 concepts into two dimensions using UMAP on the sparse concept codes, coloring each point by its energy difference $\delta(\cdot)$ so that suppressed and exaggerated blindspots stand out. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of every model pair, rendered as both scatter plots and a correlation matrix. These correlations put a single number on how much each pair of models agrees, distinguishing universal blindspots from model-specific ones, and reveal whether blindspots cluster in concept space.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the two extremes of the ranking. Suppressed blindspots are drawn from the low tail ($\delta(\cdot) < 0.1$) and exaggerated blindspots from the high tail ($\delta(\cdot) > 0.9$). Presented examples are manually annotated with textual concept descriptions obtained by inspecting each concept’s most activating images and spatial attention patterns. We outline ongoing efforts to automate this concept interpretation in Appendix D.

Section 5.5 We run a controlled A/B comparison of SD 1.5 with and without DPO. For each image pair (x, x') we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before averaging. This gives a single per-datapoint divergence score measuring how far each generated image departs from its natural counterpart in concept space. Overlaying the two variants’ score histograms then makes the shift in median and spread directly readable.

Section 5.6 For each concept c_k we count its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ (the number of non-zero activations across the natural dataset) and apply a sigmoid normalization with temperature $T = 0.4$ to the energy differences $\delta(\cdot)$. We then plot frequency (x-axis) against the sigmoid-transformed energy difference (y-axis), with point size proportional to activation frequency and color proportional to the magnitude of the energy difference. This visualization makes the frequency–fidelity relationship—frequent concepts near balanced, rare concepts strongly diverging—directly legible in every model.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a single scalar of conceptual divergence per pair. Ranking all pairs by this score places the two failure modes at opposite ends: the lowest-scoring pairs flag potential memorization artifacts, while the highest-scoring pairs flag significant conceptual failures. Reading either extreme of the ranking thus isolates a distinct failure mode for qualitative inspection.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data, D_G and D_{g_θ} , we analyze concept co-occurrence through the co-activation matrix $Z^T Z$, which records pairwise correlations in concept usage. We then eigendecompose these matrices to extract the dominant conceptual directions and measure the alignment between natural and synthetic structure via cosine-similarity heatmaps between the top-100 eigenvectors of each. In these 100×100 heatmaps a perfectly diagonal pattern would signal identical principal concept axes (a similarity of 1 along the diagonal), whereas off-diagonal mass quantifies rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56], the 32,000-concept dictionary that sets the resolution of every measurement in this paper. The model is open-sourced at a software repository.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB images). Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed to the SAE without class or position embeddings. Over 50 epochs this amounts to a substantial training budget of $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ tokens.

Dictionary. The dictionary spans 32,000 concept dimensions—to our knowledge the largest such RA-SAE to date. The sparse activation rule is top- k masking with $k = 5$: all but the five largest activations per input are set to 0. Weights are initialized with Xavier/Glorot, and training runs at mixed precision (fp16) with the final ten of the 50 epochs at full precision.

Optimizer and Schedule. The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

1009 C Computational Resources

1010 This section summarizes the GPU resources used for training and experiments in support of this
1011 paper. In total, the full study—dictionary training plus all four models’ generation and analysis—cost
1012 approximately 202 GPU-hours on NVIDIA H100s and H200s, a modest budget for the granularity of
1013 measurement obtained.

1014 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

1015 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
1016 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately
1017 100 GPU-hours on a NVIDIA H200.

1018 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
1019 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

1020 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
1021 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
1022 H200 GPU (total ~ 15 GPU-hours).

1023 D Autointerpretability

1024 In an effort to automate the interpretation of the 32,000 concepts identified by the RA-SAE, we
 1025 qualitatively evaluated how well Vision-Language Models (VLMs) can describe them. Specifically,
 1026 we queried ChatGPT 4o via the OpenAI API with various prompts and formats of an image x'' in
 1027 which the target concept was salient, recording each generated description. This section reports what
 1028 we found. For evaluation we let the VLM output its full reasoning; a one-line prompt change—asking
 1029 for the description only—would strip the reasoning to make the procedure scalable.

1030 First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superim-
 1031 posed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

1032 As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of
 1033 describing the wood texture in the background—the actual salient region—it described the ring in the
 1034 foreground, which received no attention from the RA-SAE.

1035 To determine whether this was a prompting issue, we tested the same scenario again—this time using
 1036 an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

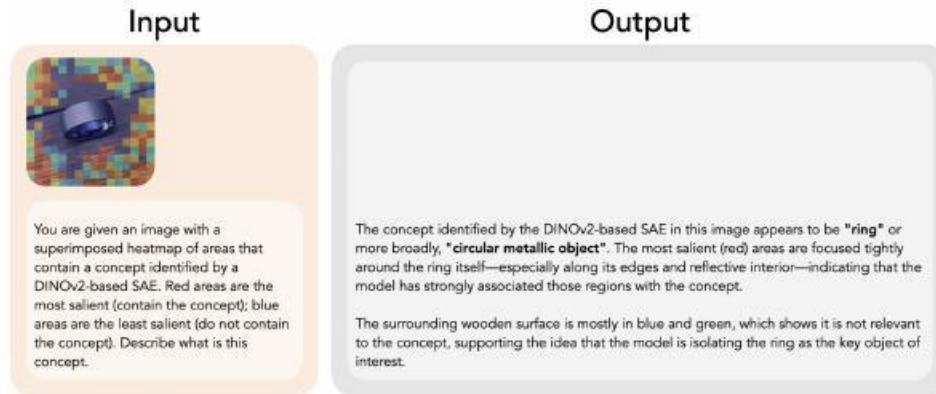
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

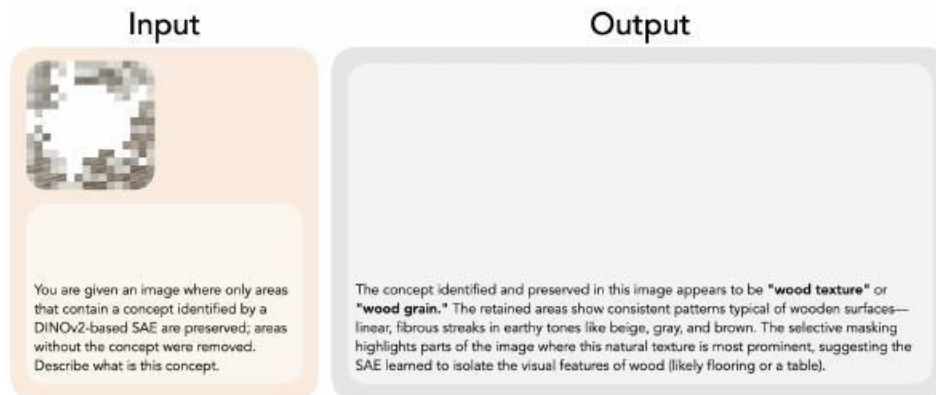
1037 This prompt modification did not resolve the issue, as the VLM still identified the ring as the
 1038 prominent concept in the image. Several other prompting techniques, including those describing the
 1039 heatmap in even greater depth (e.g., by including the code used for its generation), were explored
 1040 with similar results.

1041 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
 1042 explaining how to interpret the heatmap:



1043 Once again, the result failed to identify the wooden background as the prominent concept in the image
 1044 and instead focused on the ring. As in the previous case, we experimented with various prompting
 1045 strategies using this image format, but the results remained largely unchanged.

1046 From these observations, we concluded that the heatmap alone is insufficient to steer the VLM's
 1047 attention to the salient regions: the presence of less relevant regions consistently distracted the model.
 1048 To address this, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



1049 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 1050 and this success held consistently across a range of concepts and images.

1051 This combination of alpha-masked x'' and prompt gave the most reliable results in our qualitative
 1052 evaluation. We stress two limits on its scope: the evaluation is qualitative (there are no ground-truth
 1053 concept annotations to score against) and it covers a single VLM. We hope future work on the
 1054 autointerpretability of SAE concepts can build on and quantify this analysis.

E Additional Results: Model-Specific Blindspots

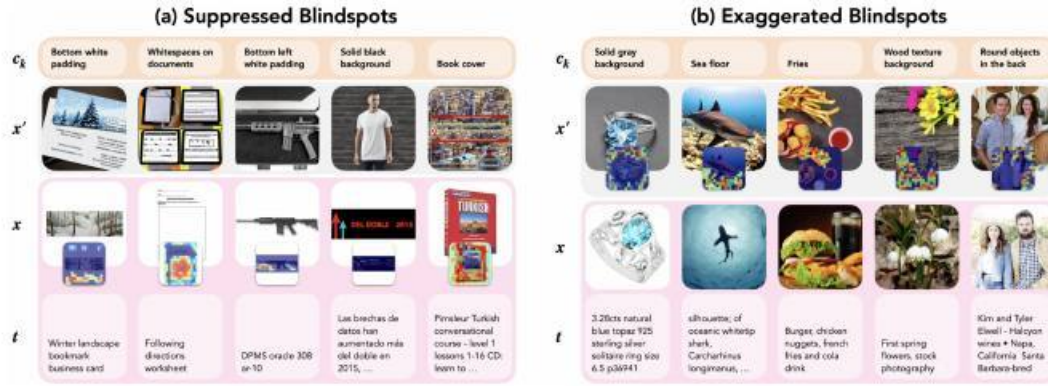


Figure 14: Concepts at the extremes of the δ score for **Stable Diffusion 1.5**. For each concept we show the single most activating image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the score is driven.

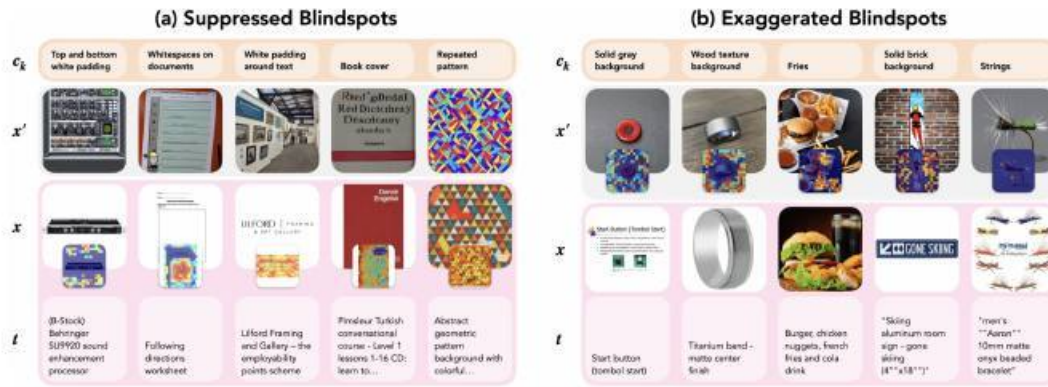


Figure 15: Concepts at the extremes of the δ score for **Stable Diffusion 2.1**. For each concept we show the single most activating image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the score is driven.

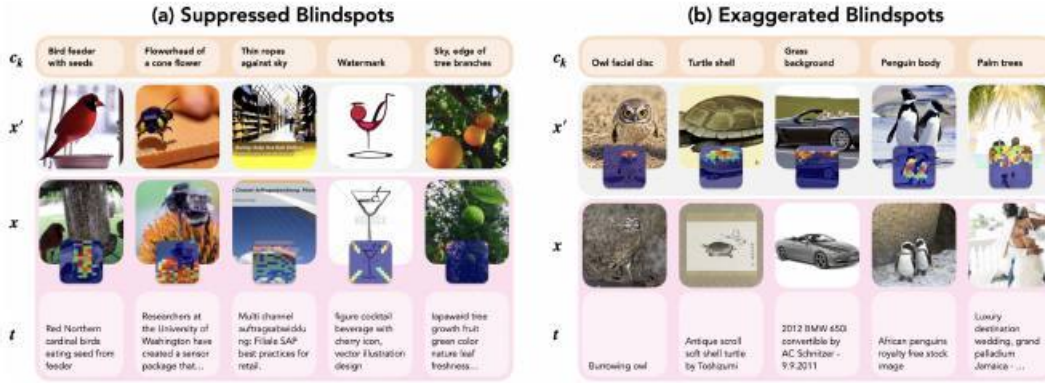


Figure 16: Concepts at the extremes of the δ score for **Kandinsky**. For each concept we show the single most activating image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the score is driven.



Figure 17: Concepts at the extremes of the δ score for **PixArt**. For each concept we show the single most activating image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed to localize where the score is driven.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far our analysis has scored individual concept activations. Yet visual scenes are rarely built from isolated concepts; they are organized by rich co-occurrence patterns that encode compositional semantics. We now measure whether generative models capture this higher-order structure, analyzing the co-activation matrix $Z^T Z$ of pairwise correlations in concept usage and comparing it, quantity by quantity, between the natural and synthesized distributions.

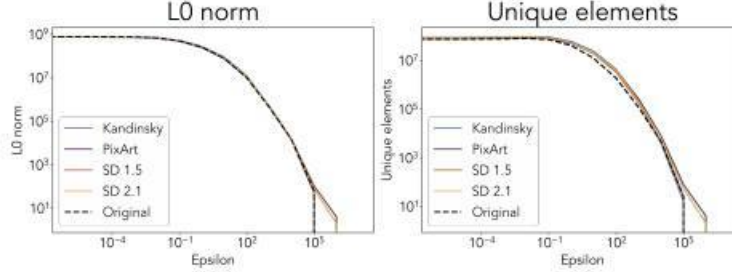


Figure 18: • **Sparsity and Structural Divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , i.e., how many entries stay active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural one. The two panels tell a two-part quantitative story: all four models track the natural global sparsity closely (left), yet diverge substantially in *which* pairs are active (right).

Surprisingly, at the level of binary structure diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded across a range of ϵ values—tracks the natural distribution closely for all four models. Quantitatively, then, the gross connectivity of the conceptual graph (which concepts co-activate at all) is well preserved. Formally, $Z^T Z$ is the adjacency matrix of a weighted, undirected graph over concepts whose edge weights reflect co-activation strength across the dataset.

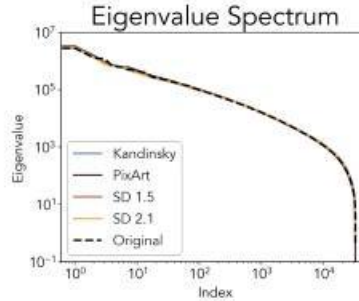


Figure 19: • **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and the natural distribution. All four models trace the same heavy-tailed, near power-law decay as the natural distribution, evidence that second-order variance allocation is quantitatively preserved.

However, as Fig. 18 (right) makes explicit, the *content* of these co-activations diverges: a substantial fraction of the entries in each model-generated $Z^T Z$ has no counterpart in the natural baseline. So while the capacity for compositionality is retained in aggregate, the identity of the active pairings shifts measurably—a signature of model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 plots the eigenvalue spectra for each model and the natural distribution. Every spectrum exhibits the same heavy-tailed, near power-law decay, indicating that the models preserve the overall rank structure and variance allocation across conceptual dimensions.

Further, we quantify the alignment of dominant conceptual directions via cosine-similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20).

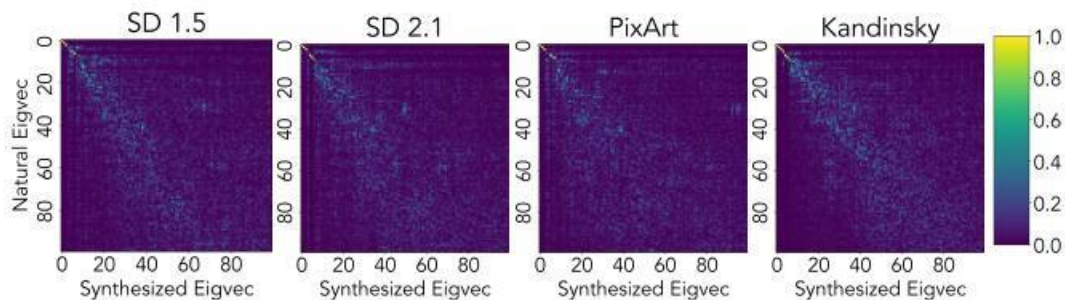


Figure 20: ● **Concept Basis Similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal mass measures alignment of the dominant conceptual directions; the strength of that diagonal varies across the four models, giving a per-model readout of how faithfully each preserves the principal concept axes.

1080 All models show partial diagonal alignment—overlap in the leading principal axes—but the off-
 1081 diagonal mass grows in the higher modes, quantifying rotations and mismatches that accumulate as
 1082 compositional order increases.

1083 Together, these measurements paint a consistent picture: diffusion models reproduce the global
 1084 shape of concept co-activation—sparsity and spectrum—remarkably well, yet deviate in structured,
 1085 quantifiable ways in both the identity of active pairings and the higher spectral modes. These
 1086 higher-order discrepancies plausibly underpin failures in generating coherent multi-object scenes or
 1087 relational concepts.

1088 G Additional Results: Blindspot Stress Testing

1089 To stress-test the blindspots our method identifies, we gathered many prompts describing each
 1090 blindspot and generated images from them, then set up a direct contrast: outputs from a model in
 1091 which the concept scored as a blindspot versus outputs from a model in which it did not. This turns
 1092 each identified blindspot into a falsifiable prediction about which model should fail.

1093 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1094 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1095 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1096 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by
 1097 the intensity with which the desired concept appeared. All images were then manually reviewed to
 1098 determine whether the blindspot was successfully depicted.

1099 As the following examples of suppressed concepts show, while peripheral aspects of the target concept
 1100 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1101 the glossy DVD disc blindspot), the models overwhelmingly failed to render the full concept. This
 1102 behavior matches, prompt-for-prompt, the low δ scores our method assigned to these concepts—an
 1103 independent, out-of-sample confirmation that the score predicts real generation failures.

1104 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from various prompts involving the bird feeder concept during stress testing. Our method scored this concept as a suppressed blindspot in Kandinsky but not in SD 1.5, and the generations bear this out exactly: Kandinsky fails to produce the concept while SD 1.5 renders it successfully—a clean, side-by-side confirmation of the score’s predictive validity.



Figure 22: Images generated from various prompts involving the glossy DVD disc concept during stress testing. Here the roles reverse: our method scored this concept as a suppressed blindspot in SD 1.5 but not in Kandinsky, and again the generations match—SD 1.5 fails to produce the concept while Kandinsky succeeds, confirming the score predicts failures at the level of the individual model.

1106 H Additional Examples: Synthesized Images



Figure 23: Additional LAION-5B image-caption pairs, each shown with prompt-matched generations from all four models (SD 1.5, SD 2.1, PixArt, and Kandinsky). As throughout, every model is driven by the identical caption, keeping the four-way comparison controlled.

1107 I Concentration bounds for δ

1108 In our experiments, we estimate each $\delta(k)$ from $n = 10,000$ paired samples. This raises a fair
 1109 question about the reliability of those estimates: is a sample of 10,000 enough? We answer it
 1110 quantitatively by deriving a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ via McDiarmid's
 1111 inequality [142]. The bound is tight and shows that even this modest sample size yields fast, accurate
 1112 estimates of concept bias.

1113 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1114 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1115 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1116 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1117 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1118 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1119 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1120 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1121 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1122 which proves the claim. □

1123 In practice most concept scores are sparse, with the bulk concentrated near zero and only a few
 1124 reaching values up to 10. Even at those largest observed biases, the bound guarantees that $n = 10,000$
 1125 samples keep $\hat{\delta}_n(k)$ within a small ε of its true value with high probability—the deviation shrinking
 1126 exponentially in n . This quantitatively justifies our sampling budget and confirms that accurate bias
 1127 measurements are attainable with limited data.

1128 J Monotonicity and Calibration-Free Interpretation of δ

1129 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias, and
 1130 in practice we rank by the score $\delta(k)$. One might worry that squashing the raw signal through a
 1131 sigmoid distorts this ranking relative to more direct quantities such as the energy gap or the odds
 1132 ratio. The following result rules that out: $\delta(k)$ is a strictly increasing reparameterization of both and
 1133 therefore induces exactly the same ordering. Consequently, no calibration is needed—the score can
 1134 be compared across concepts and models at face value.

1135 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1136 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1137 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1138 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1139 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1140 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1141 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1142 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1143 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1144 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1145 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1146 concepts.

1147

□

1148 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by the energy gap or by the
 1149 conceptual generation odds ρ_k : all three impose the identical total order. No calibration is necessary,
 1150 and the bounded $[0, 1]$ score can be read directly as a severity measure.

1151 K Stability of FID Under SAE Embeddings

1152 In this section we establish a tight, two-sided quantitative bound relating the Fréchet Inception
 1153 Distance (FID) computed in the original activation space (of dimension d) to the FID computed
 1154 after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we
 1155 assume that D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1156 We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an
 1157 isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not
 1158 realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme
 1159 singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$
 1160 specialised to Gaussians.

1161 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
 1162 $\mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently
 1163 the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1164 Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a
 1165 simple lemma in the case where D is not overcomplete.

1166 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1167 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1168 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1169 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
 1170 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical.
 1171 □

1172 This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract
 1173 meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but
 1174 we can still have column-orthonormal dictionary. We will use that to show that we can bound using
 1175 the extremal singular value of $D^\top D$.

1176 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1177 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1178 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1179 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1180 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1181 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1182 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1183 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1184 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1185 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1186 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1187 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1188 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1189 finishes the proof. $\square$

1190 In short, Theorem 6 bounds the distortion: applying a column-orthogonal overcomplete SAE dictio-
 1191 nary can change the Fréchet Inception Distance by at most a factor equal to the square of its extremal
 1192 singular values. Since we empirically find $\sigma_{\min}, \sigma_{\max} \approx 1$, both multiplicative bounds collapse
 1193 toward 1, so FID measured in the SAE feature space is essentially identical to the canonical FID—the
 1194 embedding preserves the metric almost exactly. $\square$

1195